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Assignment 07

Document Question Answering System using Streamlit and LangChain

1. Objective

The objective of this assignment is to build a Document Question Answering (QA) system using Streamlit for the user interface and LangChain for backend processing. The system allows users to upload documents and ask questions based on the content of those documents.

2. Tools and Technologies Used

- Python
 - Streamlit (Frontend UI)
 - LangChain (RAG Pipeline)
 - ChromaDB / FAISS (Vector Database)
 - HuggingFace Embeddings
 - LLM (Ollama / OpenAI / Local Model)
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3. System Overview

The system is designed as a Retrieval-Augmented Generation (RAG) pipeline where:

- User uploads a document (PDF / TXT)
 - Document is processed and split into chunks
 - Embeddings are generated for each chunk
 - Chunks are stored in a vector database
 - User asks a question via Streamlit UI
 - Relevant chunks are retrieved
 - LLM generates final answer based on retrieved context
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4. Architecture Flow

The system follows this workflow:

1. **Document Upload (Streamlit UI)**
 2. **Text Extraction from File**
 3. **Text Splitting (Chunking)**
 4. **Embedding Generation**
 5. **Vector Store Creation**
 6. **Similarity Search (Retrieval)**
 7. **LLM-based Answer Generation**
 8. **Display Answer in UI**
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5. Implementation Details

5.1 Streamlit UI

- Provides interactive web interface
 - Allows file upload
 - Accepts user queries
 - Displays final answers
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5.2 LangChain Pipeline

LangChain is used to connect all components:

- Document Loader (PyPDFLoader / TextLoader)
 - Text Splitter (RecursiveCharacterTextSplitter)
 - Embedding Model (HuggingFace / SentenceTransformers)
 - Vector Store (ChromaDB / FAISS)
 - Retrieval Chain (QA Chain)
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5.3 Question Answering Process

1. User enters query
2. Query is converted into embedding

3. Similar document chunks are retrieved
 4. Retrieved context is passed to LLM
 5. Final answer is generated
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6. Key Features

- Upload and query any document
 - Real-time question answering
 - Context-aware responses using RAG
 - Simple and interactive UI using Streamlit
 - Efficient retrieval using vector database
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7. Advantages

- Easy-to-use interface
 - Fast and efficient document search
 - Works with large documents
 - Improves answer accuracy using retrieval context
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8. Limitations

- Performance depends on embedding quality
 - Large documents may slow retrieval
 - Requires proper chunk tuning for best results
 - LLM hallucination may occur without strong context
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9. Results

- The system successfully answers questions based on uploaded documents
- Retrieval improves accuracy compared to direct LLM prompting
- Streamlit UI provides smooth user interaction
- LangChain ensures modular and scalable architecture

10. Output:



11. Conclusion

In this assignment, a Document Question Answering system was successfully developed using Streamlit and LangChain. The integration of RAG architecture significantly improves the accuracy of responses by combining retrieval and generation. This system demonstrates a practical use case of modern LLM-based applications for document understanding.